

Assignment

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
 2. Your answer should be **handwritten**
 3. Submit online here: <https://forms.gle/FkXyUUP3WHUNFWeg9>
 4. **NO LATE SUBMISSION WILL BE ACCEPTED.**
 5. Check the **section deadline mentioned** in discord.
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Q1. [8 Marks] For the given table titled **Characters**

ID	Character	Anime	Role
N-107	Naruto Uzumaki	Naruto	Protagonist
DN-245	Light Yagami	Death Note	Protagonist
AOT-389	Mikasa Ackerman	Attack on Titan	Soldier
AOT-512	Levi Ackerman	Attack on Titan	Captain
OP-736	Monkey D. Luffy	One Piece	Pirate
AOT-821	Erren Yeager	Attack on Titan	Soldier
DBZ-403	Goku	Dragon Ball Z	Fighter
FMA-659	Edward Elric	Fullmetal Alchemist	Alchemist
CB-914	Spike Spiegel	Cowboy Bebop	Bounty Hunter
DBZ-208	Vegeta	Dragon Ball Z	Prince

Make a hash index on the attribute **ID**. Suppose the number of buckets in the hash index is 4 and each bucket can house 2 index entries at max. Bucket overflow is handled using forward chaining.

The hash function, $h = (\text{sum of all digits in ID}) \% 4$

example: $h(N-107) = (1+0+7) \% 4 = 0$

Only draw the buckets (with forward chaining, when applicable). No need to connect with the original table.

Q2. [12 Marks] A B+ Tree of order $n = 3$ is provided below.

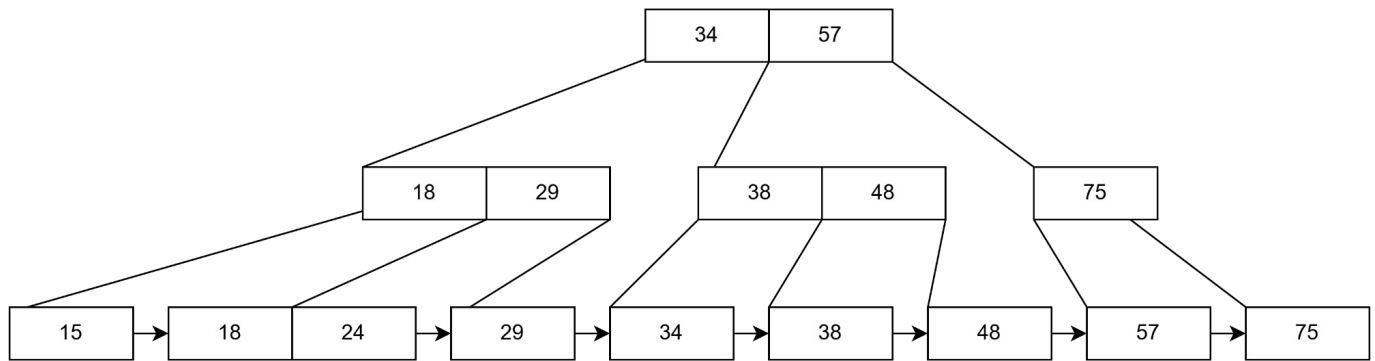


Figure: A

Do the following operations on the given tree (Figure: A) sequentially.

- Delete 18
- Delete 24
- Delete 75
- Delete 57
- Delete 38
- Delete 29

You need to draw the tree for each step after deletion.
